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# Experimental and numerical estimation on roll damping and pressure on a 2-D rectangular structure in free roll decay test

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ABSTRACT

Experimental and numerical studies were performed to estimate roll damping and hydrodynamic pressure on a 2-D rectangular structure. Free roll decay test was simulated by experiment in 2-D wave tank with Particle Image Velocimetry (PIV) measurement, and by numerical simulations using CFD based on an unsteady incompressible Reynolds-Averaged Navier-Stokes (RANS) approach. The dynamic characteristics of the roll motion and the roll damping coefficients from the experiment and CFD were analyzed and compared. Velocity and vorticity fields obtained from the PIV measurement were examined to understand the fluid fields in the vicinity of the structure and compared with them from CFD to validate the numerical methods applied in the study. The PIV-based pressure estimation method was applied to predict pressures on the structure, and results were compared with them from CFD to assess the feasibility of the PIV-based estimation approach. The overall agreement between experimental and CFD results was good for roll damping coefficient, velocity and vorticity fields, and pressure considering the complexity of fluid physics in the roll motion. It was found that the PIV-based pressure estimation method would be alternative technique to estimate the pressure and loads on the structure.

## 1. Introduction

Accurate estimation of the roll motion for floating structures is crucial to the hull design, connection of the risers, topside structures and process plants on board ultimately to ensure the comfortability of crews, the dynamic stability and operability of the structures. Unlike the other degrees of dynamic motions in response to the environmental loads, the roll motion is highly nonlinear and sensitive to the hull shape of the structures due to the fluid viscosity. It is a difficult task to estimate the roll damping because of the complexity to quantify the viscous effect which is known to have significant influence on the roll motion of a blunt shaped floating structure.

There have been many researches to solve the roll motion issues efficiently using empirical, numerical and experimental methods. The early works to predict the roll damping and motion were done using semi-empirical formulae derived from experiments and hydrodynamic theory. Empirically, Ikeda et al. (1977) proposed a formula in which the eddy making component of the roll damping force for a 2-D body is proportional to the square of the frequency. They deduced the amplitude of the roll motion from the semi-experimental technique based on theoretical and experimental results. Chakrabarti (2001) examined the roll damping components and their empirical contributions from a series of model experiments. He decomposed the components for a ship hull form to the skin friction of the hull, eddy shedding from the hull, free surface waves, lift effect damping, and bilge keel damping including the consideration of scale effects.

Numerically, potential flow theories are still widely used to assess the hydrodynamic performance of ships and offshore structures even though potential-based methods cannot predict the roll motion issues efficiently due to the negligence of fluid viscosity and rotation. Salvesen et al. (1970) presented that potential flow methods overestimated roll amplitude responses of rectangular structures in beam seas compared to results from experiments. Empirical factors are commonly introduced to predict the roll damping coefficient taking into account the viscous effects due to flow separation and generation of vortices (Downie, 1987). Recently computational fluid dynamic (CFD) methods have been

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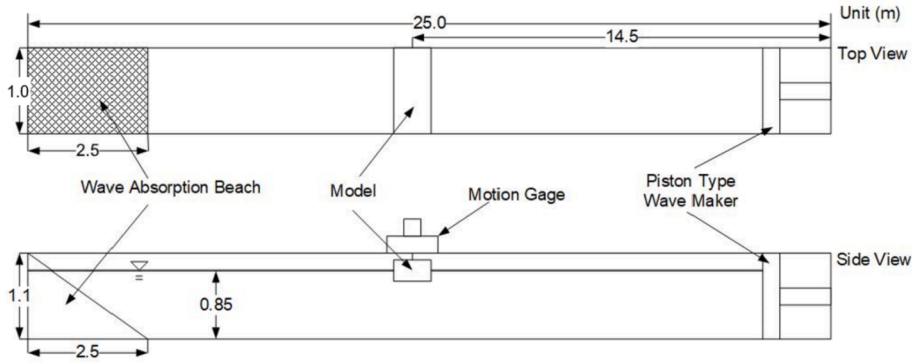



**Fig. 1.** 2-D wave tank.

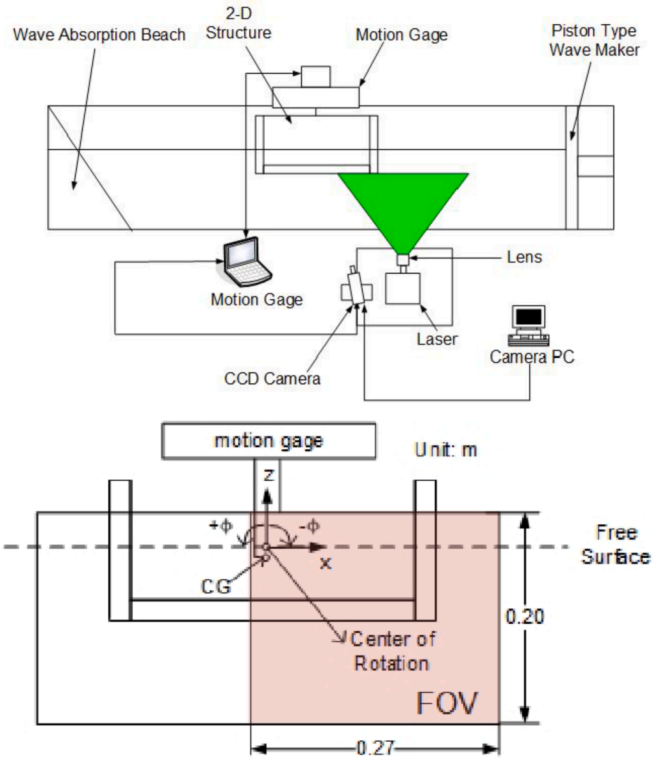



**Fig. 2.** Experimental setup in 2-D wave tank (top: experimental schematics, bottom: FOV of PIV).

increasingly applied to study the roll damping considering the viscous effects and to investigate the vortex-induced phenomena such as vortex shedding. Chen et al. (2001) employed a Reynolds-Averaged Navier--Stokes (RANS) method with a chimera domain decomposition approach to simulate large roll motions. Wilson et al. (2006) demonstrated the viscous phenomenon of roll decay motion of a surface combatant using the RANS CFD code, CFDSHIP-IOWA. Chen et al. (2016) computed the roll motion of a 2-D rectangular structure in regular waves and compared the velocity and vorticity fields in the vicinity of the structure using an extended OpenFOAM (open-source CFD package) model. However, it has been reported that there is still a shortage in CFD simulations and more computational results are needed to establish a general approach to estimate the roll motion and viscous effects more accurately (Irkal et al., 2016; Gokce and Kinaci, 2018).

Experimentally, there have been technical advances in the recent years especially for fluid measurement. Particle image velocimetry (PIV) has been used as a reliable and versatile technique to measure fluid fields and to examine the fluid physics, especially for the structures of vortices ruling the dynamics of systems in an efficient and non-intrusive manner. Yeung et al. (1996) measured the vorticity fields generated by a plate undergoing a forced harmonic roll motion at free surface using the PIV technique. They also conducted a numerical simulation to reproduce the observed flow pattern of the rolling rectangular cylinder using the random vortex method with the presence of free surface (Yeung et al., 1998). Images were taken from the velocity field around the barge to capture the behavior of the fluid and to understand the roll damping. Dong et al. (1997) conducted PIV measurements to study the vorticity structure near the bow of a ship model due to surface waves. Jung et al. (2006) obtained flow maps using the PIV in a 2-D wave tank to elucidate the vertical flow characteristics in vicinity of a rectangular barge. Aloisio and Felice (2006) investigated the vorticity field evolution around the bilge keels of a ship model in free roll decay motion to understand the interaction between the vortices and the hull. Oliveira and Fernandes (2014) studied nonlinear damping behavior of the Floating Production Storage and Offloading (FPSO) with bilge keels and vortex patterns for the interaction between the vortices created during the roll motions.

Regarding the prediction of pressure distributions and overall forces on offshore structures, designers generally use velocities obtained from model experiments. However, the estimation of hydrodynamic pressure and loading on the structures is a major scientific and engineering challenge and still open question due to a very high pressure and a short duration (Ariyarathne et al., 2012). The evaluation of the hydrodynamic loading is currently relying on the direct measurement of the pressure using pressure gauges mounted on the structures, but it is difficult to install the gauges and limited to local measurements. In aerodynamic practice, the PIV-based pressure estimation approach has been applied as an attractive and feasible method to examine the pressure and loads using flow field information from PIV measurement (van Oudheusden, 2013), even when model instrumentation is problematic (e.g. icing on airfoils (De Gregorio, 2006)); the observation of freely flying birds or insects (Spedding and Hedenström, 2009). In the research of the ocean engineering, Jakobsen et al. (1997) used PIV to determine acceleration and pressure in wave phenomena. Panciroli and Porfiri (2013) applied the pressure-gradient integration using the PIV data for a rigid wedge entering a water surface and verified the estimated pressure with Wagner's solution based on potential flow theory. Their findings demonstrated the accuracy of the reconstruction of pressure field over the entire impact duration.

In this paper, experimental and numerical studies were performed to investigate the roll damping and estimate pressure on a 2-D floating rectangular structure for the roll free decay in roll and heave free condition. Firstly, the dynamic characteristics of the roll motion were investigated to predict the roll damping coefficients by experiment in 2-D wave tank and numerical simulations based on an unsteady incompressible RANS method for the CFD. Grid convergence test for the CFD was conducted to capture the wave propagation from the structure properly. Secondly, the velocity and vorticity fields were examined to

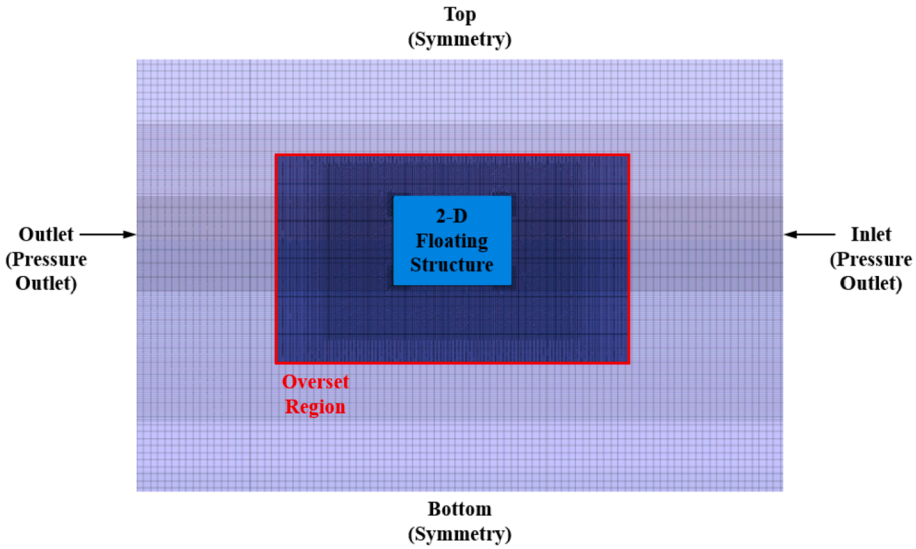



**Fig. 3.** Grid and boundary conditions.

understand the fluid fields in the vicinity of the structure in the roll motion, and the experimental and numerical results were compared especially to validate the numerical methods applied in the study. Thirdly, estimation results of the pressure and forces on the structure using velocity field information from PIV measurement were presented with explanation of the PIV-based pressure estimation approach. Finally, pressure and forces on the structure from experiment and CFD were compared to validate both the PIV-based pressure estimation and CFD methods.

## 2. Experimental setup

The experiment was carried out for the roll free decay test of a rectangular structure in a 2-D wave tank with a piston type wave maker driven by a synchronous servo-motor controlled by a computer. The tank is 25 m long, 1 m wide, and 1.1 m deep glass-walled flume with a permeable wave absorbing beach of 2.5 m length and 1.1 m height at the end of tank to minimize reflection as depicted in Fig. 1.

A rectangular acrylic structure with the dimensions of 0.20 m wide, 0.99 m long, and 0.15 m high was located at 14.5 m from the right end of the tank. The water depth of the tank was 0.85 m throughout the experiments. The structure was mounted to the tank with bars and a pair of hinges adjusted to the axis aligned with the calm water level as displayed in Fig. 2. The draft ($D$) and center of gravity (CG) of the structure are 0.07 m and 0.06 m from the keel respectively. The hinges of the motion gage allowed the structure to roll and heave motions and aligned with water level but restrained it from surge and sway motions. The sampling rate was set at 100 Hz and test results of motions were analyzed without any filtering technique. The error analysis for the measurement using the motion gage was conducted before measuring the motions in the experiment. The errors of linearity, hysteresis and repeatability are 0.971%, 0.891% and 0.642% respectively, and the total error is 1.466%.

PIV technique was used to measure 2-D velocity field in the vicinity of the structure. The PIV system was synchronized by the camera computer and motion gage housing data acquisition boards as schematically illustrated in Fig. 2. The digital high-speed CCD camera mounted with a 105 mm f/2.8 micro focal lens was used to obtain PIV images. It had 2352 × 1728 pixels (i.e. 7 μm × 7 μm pixel size) with 8-bit dynamic range. A pair of images were obtained by double-frame/single pulsed method. The time difference between successive frames was set at 0.005s, which was determined by the maximum displacement to be less than a third of the width of the interrogation window size.

The field of view (FOV) was used to obtain velocity and vorticity maps at the right-lower corner of the structure. The FOV size were 270 × 200 $mm^2$ corresponding to the spatial resolution of 3.75 mm between every velocity vectors in the interrogation area of 64 × 64 pixels with 50% overlap. The water was seeded with small particles that have a mean diameter of 57 μm and a specific gravity of 1.02. The images were analyzed to compute PIV velocity vectors using the DynamicStudio which is the commercial software developed by DANTEC DYNAMICS. The adaptive-correlation between the two interrogation windows from a pair of images was used for the PIV velocity vector analysis.

## 3. Mathematical background

### 3.1. Computational fluid dynamics

An unsteady incompressible RANS (Reynolds-Averaged Navier-Stokes) approach was applied to calculate the roll amplitude and obtain numerical velocity and vorticity maps in roll free decay test using a commercial CFD software STAR-CCM+ (CD-adapco, 2015). For incompressible flows, if there are no external forces, the averaged continuity and momentum equations are given in tensor form in the Cartesian coordinate system by Eq. (1) and Eq. (2)

$$\frac{\partial(\rho\overline{u}_i)}{\partial x_i} = 0 \tag{1}$$

$$\frac{\partial(\rho\overline{u}_i)}{\partial t} + \frac{\partial}{\partial x_j}\left(\rho\overline{u}_i\overline{u}_j + \rho\overline{u_i'u_j'}\right) = -\frac{\partial\overline{p}}{\partial x_i} + \frac{\partial\overline{\tau}_{ij}}{\partial x_j} \tag{2}$$

where $\overline{u}_i$ is the relative averaged velocity vector of flow between the fluid and the control volume, $\overline{u_i'u_j'}$ is the Reynolds stresses and $\overline{p}$ is the mean pressure. For Newtonian fluid under incompressible flow, the mean shear stress tensor, $\overline{\tau}_{ij}$, is expressed as Eq. (3)

$$\overline{\tau}_{ij} = \mu\left(\frac{\partial\overline{u}_i}{\partial x_j} + \frac{\partial\overline{u}_j}{\partial x_i}\right) \tag{3}$$

where $\mu$ is dynamic viscosity.

The finite volume method (FVM) and the volume of fluid (VOF) method were applied to the spatial discretization and free surface capturing respectively. The flow equations were solved in a segregated manner using a predictor-corrector approach. Convection and diffusion terms in the RANS equations were discretized by a second-order upwind scheme and a central difference scheme. The Semi-Implicit Method for

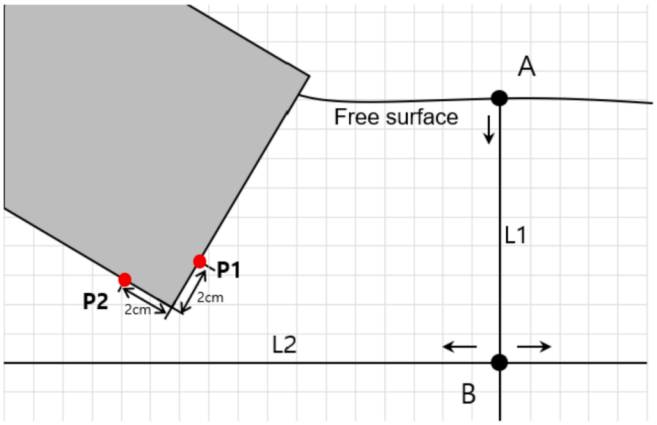



**Fig. 4.** Pressure integration scheme in roll free decay test.

**Table 1**
Test cases for grid convergence.

| Case name | Mesh | λ/Δx | H/Δz | T/Δt | no. of meshes |
|---|---|---|---|---|---|
| MC | Coarse | 100 | 22 | 333 | 452,162 |
| MM | Medium | 140 | 30 | 500 | 673,348 |
| MF | Fine | 195 | 44 | 666 | 1,582,648 |

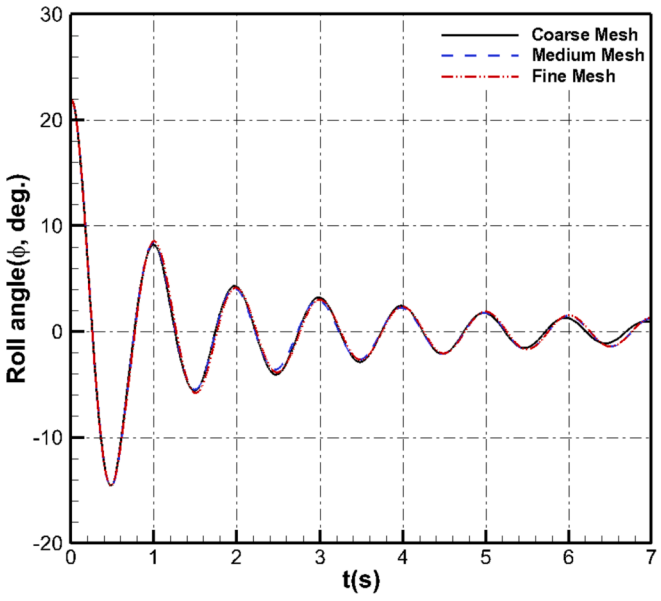



**Fig. 5.** Grid convergence test for the roll angle.

Pressure-Linked Equations (SIMPLE) algorithm was used to resolve the pressure-velocity coupling and a k-ω Shear Stress Transport (SST) model was applied as the turbulence model for all calculations. A second-order temporal scheme was used for discretization of the unsteady term in RANS equations to improve accuracy of wave propagation from the structure. In order to consider motions of the structure, a Dynamic Fluid Body Interaction (DFBI) scheme was applied with the vessel free to move in roll and heave directions.

Overset grids were used to simulate the floating structure in large amplitude motions without deforming meshes especially around free surface and the structure to make the grid quality remain the same in all simulation time. One grid is adapted to the background such as free surface, reflection waves, etc., while overlapping grids are attached to the structure and move with it without deformation.

A forcing zone was introduced to reduce the domain size and to

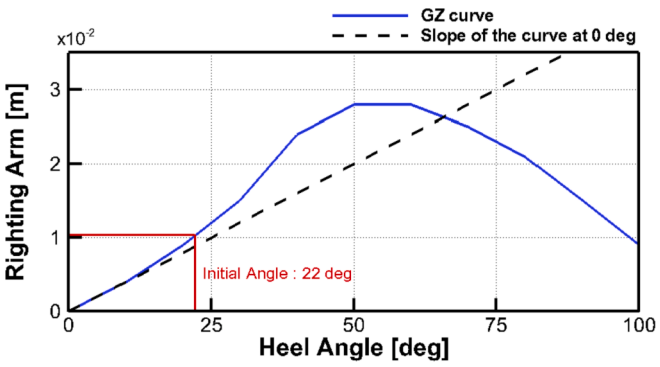



**Fig. 6.** $\overline{GZ}$-curve of rectangular structure.

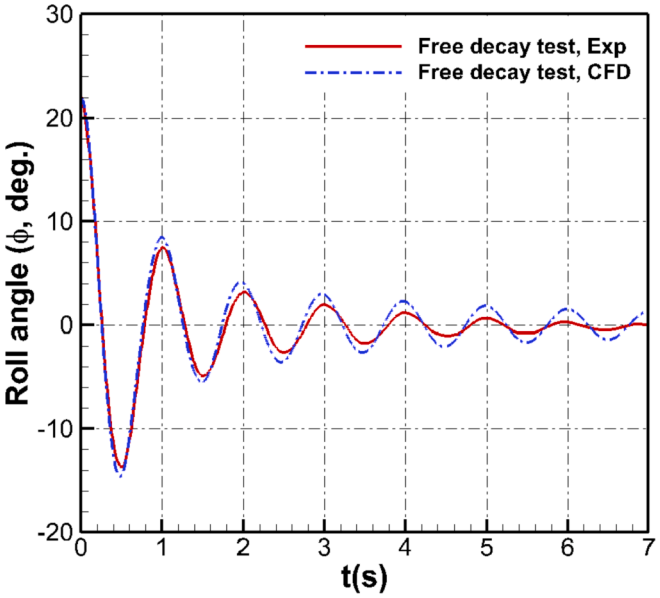



**Fig. 7.** Test of roll free decay.

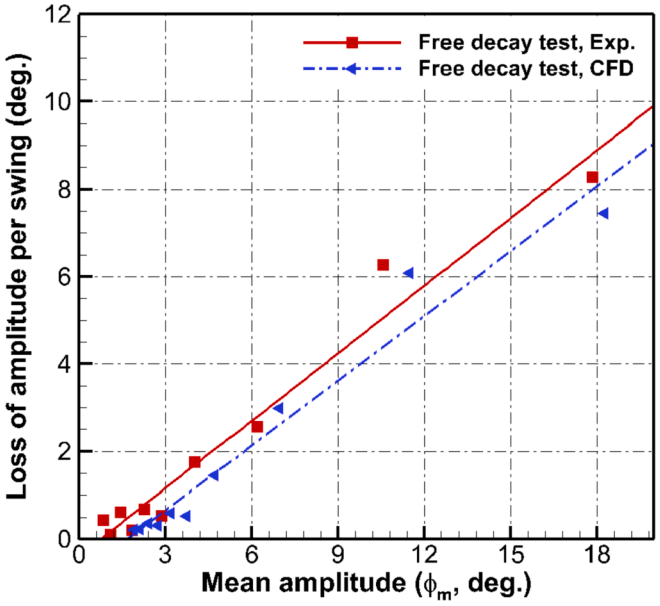



**Fig. 8.** Curve of extinction.

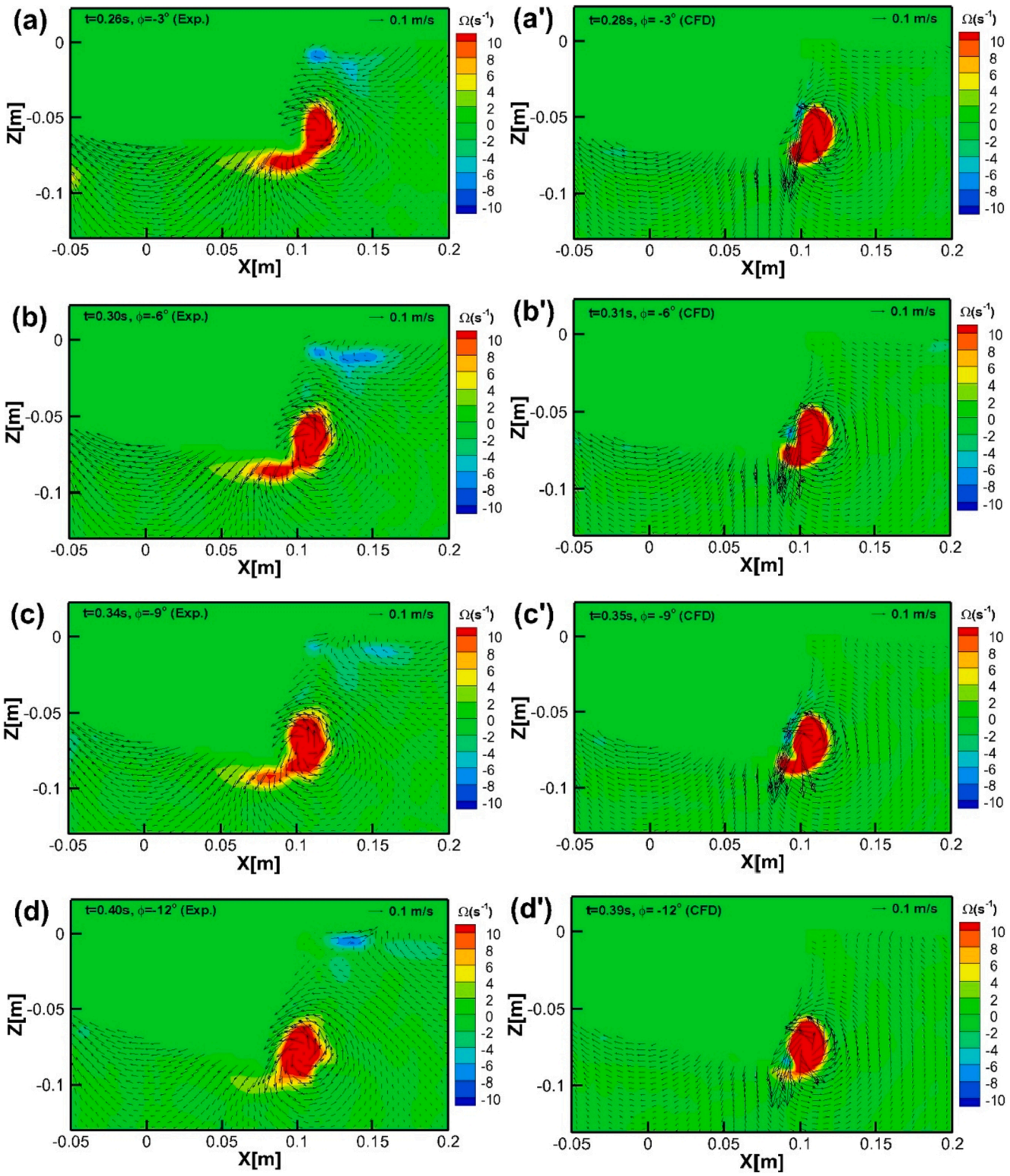



**Fig. 9.** Velocity and vorticity maps for the first negative CW roll angles of the structure (a) - (d) PIV measurements; (a') – (d') CFD simulations.

minimize reflection waves from the boundaries (Enger et al., 2014). In the zone, the disturbed flows from the structure are absorbed by actively adjusting the momentum, mass and volume fraction source. The coupled zone with flat VOF waves was implemented to eliminate disturbed flows, which were reflected from the boundaries, around the structure in the study. In this study, the forcing zone of 2 m was applied to both inlet and outlet boundaries which were set as pressure outlet condition.

Fig. 3 shows the grid refined locally in overset region around the structure and free surface. To investigate the viscous effect more accurately in the vicinity of the structure, the prism layer was applied around the structure. The calculation domain is $-6\text{ m} < x < 6\text{ m}$, $-0.85\text{ m} < z < 0.55\text{ m}$ where the mid-plane and draught of the structure are located at $x = 0$ and $z = 0$, respectively.

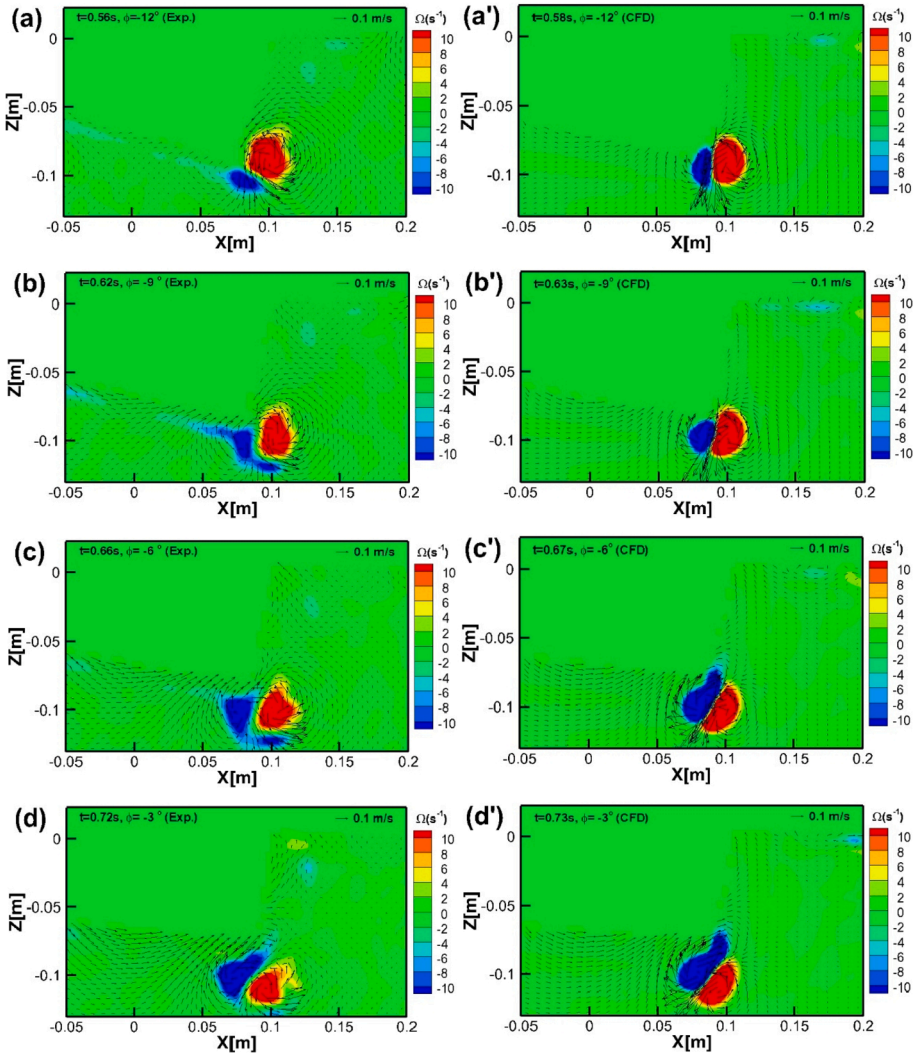



**Fig. 10.** Velocity and vorticity maps for the first negative CCW roll angles of the structure (a) - (d) PIV measurements; (a') – (d') CFD simulations.

### 3.2. PIV-based pressure estimation

The Euler equations were used to directly compute the pressure gradient components from PIV velocity data by Eq. (4) and Eq. (5)

$$\frac{\partial p}{\partial x} = -\rho\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + w\frac{\partial u}{\partial z}\right) \tag{4}$$

$$\frac{\partial p}{\partial z} = -\rho\left(\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + w\frac{\partial w}{\partial z}\right) - \rho g \tag{5}$$

where $u$ and $w$ denote the horizontal and vertical velocity components obtained from PIV technique on the axes $x$ and $z$ of a Cartesian coordinate system. The viscous terms in the momentum equation was discarded taking into account that they cannot be properly resolved from the velocity field measurements unless a sufficiently high resolution can be achieved, for which the friction contribution was located in a thin boundary layer of the structure (Jakobsen et al., 1997; van Oudheusden, 2008).

The central finite difference approximation was adopted to calculate the spatial and temporal derivatives of the velocity components as expressed by Eq. (6) and Eq. (7)

$$\frac{\partial u}{\partial x}(x,z,t) = \frac{u(x+h,z,t) - u(x-h,z,t)}{2h} \tag{6}$$

$$\frac{\partial u}{\partial t}(x,z,t) = \frac{u(x,z,t+\delta t) - u(x,z,t-\delta t)}{2\delta t} \tag{7}$$

where $h$ is the grid spacing and $\delta t$ is the time difference between two successive velocity fields. In the study, $h$ is equal to $\delta x$ and $\delta z$ as the spatial resolution of 3.75 mm in the PIV measurement. $\delta t$ was determined as 6 times (i.e. 0.03s) of the time difference between the PIV frames considering the accuracy and error accumulation of the pressure computation because increasing the time separation increases the truncation error, but reduces the effect of the velocity error on the acceleration determination (Jakobsen et al., 1997; van Oudheusden, 2013).

The forward or backward differences were employed depending on the location to calculate the spatial derivative of points at the boundary by Eq. (8) and Eq. (9) respectively,

$$\frac{\partial u}{\partial x}(x,z,t) = \frac{u(x+h,z,t) - u(x,z,t)}{h} \tag{8}$$

$$\frac{\partial u}{\partial x}(x,z,t) = \frac{u(x,z,t) - u(x-h,z,t)}{h} \tag{9}$$

Regarding the calculation of the pressure inside the fluid domain, the pressure gradient is spatially integrated from a boundary condition set at the free surface. Fig. 4 depicts the main steps of the integration procedure. Forward integration was used along vertical line L1 starting

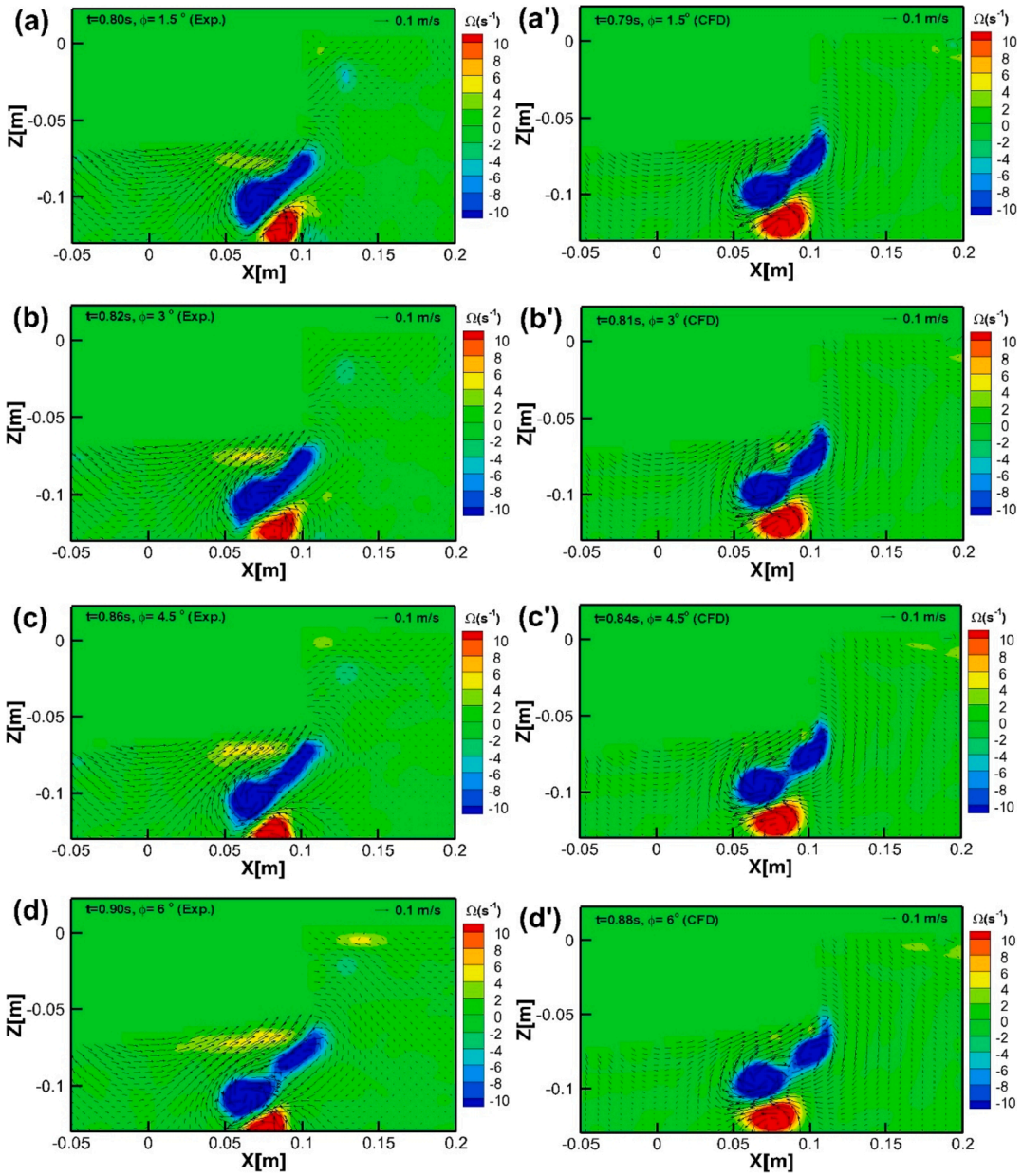



**Fig. 11.** Velocity and vorticity maps for the first negative CCW roll angles of the structure (a) - (d) PIV measurements; (a') – (d') CFD simulations.

from point A initially selected at the free surface where the pressure is set to zero. A point B on line L1 was selected to estimate all pressure along horizontal line L2 by performing forward and backward integration. From the known pressure on lines L1 and L2, the space-marching integration method proposed by Baur and Köngeter (1999) was applied to compute the unknown pressure over the rest of the fluid domain. The same procedure was performed for different points of A and B in fluid domain. The final pressure field was obtained by taking the average of all the resulting pressure fields. Points P1 and P2 were set at 2 cm away from the corner on the right and bottom wall of the structure respectively to analyze and compare pressure obtained from the PIV-based estimation and CFD methods.

## 4. Discussion of experiment and numerical results

### 4.1. Grid convergence test

Prior to the investigation on the roll damping and pressure on the structure, grid convergence tests for the CFD method were performed to capture the wave propagation from the structure properly in the roll free decay test. 0.044 m of wave height ($H$) was assumed as about 4 times ($\pm 2$ times) of $z$-position of the free surface when the structure is rotated for 22° of the initial roll angle. 1.56 m of the wave length was applied considering the roll natural period of $T_N = 1.0$ s (Kim et al., 2019).

For all test cases of grid convergence, more than 80 cells per wave length was applied based on the guidelines for ship CFD applications from ITTC (2011). A minimum of 20 cells per wave height were taken into account (Enger et al., 2014). The coarse and fine mesh systems were derived by reducing and increasing cell numbers using a factor of $\sqrt{2}$ (Bøckmann et al., 2014). The time step of the simulation was set to be proportional to the grid size as shown in Table 1.

The results of the grid convergence tests with three different mesh systems are presented in Fig. 5 where the results using medium and fine mesh are almost identical in time domain. The grid uncertainty analysis was conducted for both the roll natural frequency and roll peak angles in 7 cycles of the roll motion using grid triplets $G_1$, $G_2$ and $G_3$ with a

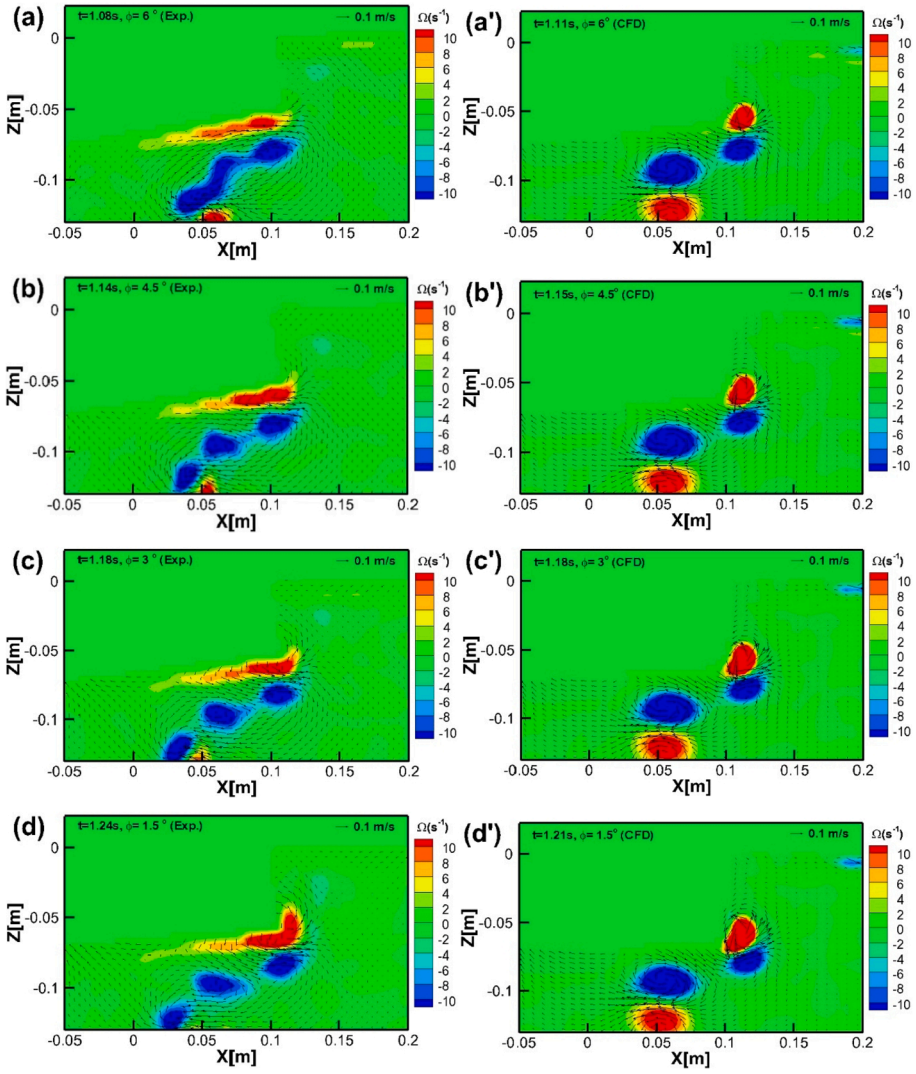



**Fig. 12.** Velocity and vorticity maps for the first negative CW roll angles of the structure (a) - (d) PIV measurements; (a') – (d') CFD simulations.

uniform parameter ratio ($R_G$) chosen as $\sqrt{2}$ for the free surface refinement (Kim et al., 2017). $S_1$, $S_2$ and $S_3$ are corresponding solutions using the fine, medium and coarse mesh, and $R_f$ and $R_p$ are the convergence ratios for the roll natural frequency and roll peak angle respectively as given in Eq. (10).

$$R_{f,\ p} = \frac{\varepsilon_{f,p21}}{\varepsilon_{f,p32}} \tag{10}$$

where $\varepsilon_{f,p21} = S_{f,p2} - S_{f,p1}$ and $\varepsilon_{f,p32} = S_{f,p3} - S_{f,p2}$ are the differences between medium-fine and coarse-medium solutions of the roll natural frequency and roll peak angle. The grid uncertainty studies showed monotonic convergences for the roll natural frequency with $R_f = 0.047$ and for the roll peak angle with $R_p = 0.183$. Therefore, the medium mesh system was chosen for the CFD simulations in the study.

### 4.2. Roll motion characteristics for 2-D rectangular structure

The dynamic characteristics of the roll motion of the 2-D rectangular structure were investigated before examining the roll damping and pressure on the structure from PIV measurement and numerical study.

Fig. 6 shows the $\overline{GZ}$-curve of the rectangular structure which was estimated to find if the rolling has as a linear or nonlinear characteristic with the initial angle of 22°. Based on the $\overline{GZ}$-curve in Fig. 6, the roll period is considered as a constant in this paper because of the small differences between the righting arm and the slope tangential at zero deg.

In this study, free roll decay tests in calm water were conducted to obtain the natural roll frequency ($\omega_N$) and the damping coefficient ($b$) experimentally and numerically. Note that the roll and heave motion were allowed in the present study to investigate the roll motion characteristics accurately (Kim et al., 2019). The equation of roll motion in free roll decay can be expressed based on the GZ-curve (Fig. 6) as below,

$$\frac{d^2\phi}{dt^2} + 2\zeta\omega_N\frac{d\phi}{dt} + \omega_N^2\phi = 0 \tag{11}$$

where $\zeta$ = damping factor; $2\zeta\omega_N = b/I'$; $\omega_N^2 = \Delta\overline{GM}/I'$. The roll motion decayed after each rolling cycle due to the damping effect as shown in Fig. 7. The natural roll frequencies ($\omega_N$) obtained from the experiment and CFD through the spectrum analysis were almost identical as 6.264 rad/s (or natural roll period $T_N$ = 1.003s) and 6.274 rad/s (or $T_N$ = 1.001s) respectively.

Fig. 8 shows the decrease of inclination for each single roll plotted against the mean angle of roll ($\phi_m$), which is called the curve of roll extinction. The formula was applied to obtain the roll damping coefficient $b$ following Bhattacharyya (1978) as below,

$$b = \frac{K_1 T_N \Delta\overline{GM}}{\pi^2} \tag{12}$$

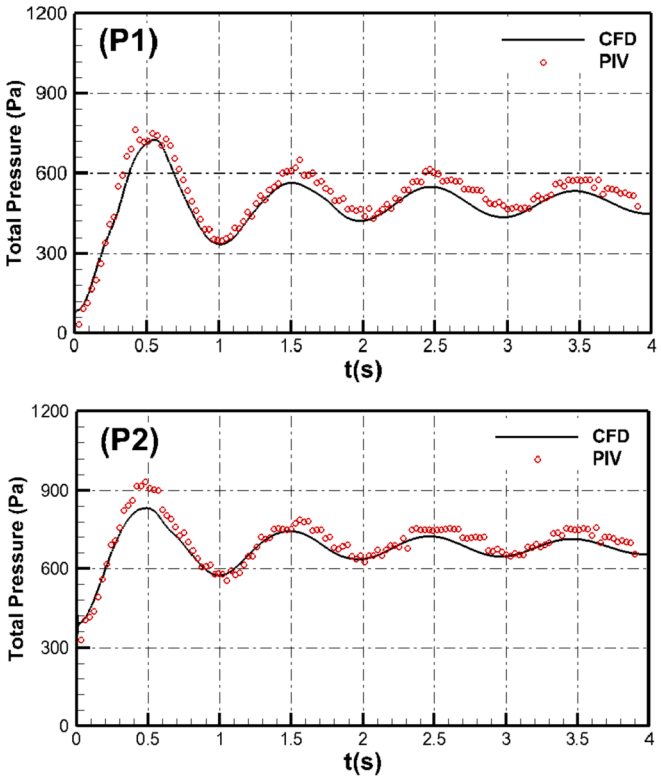



**Fig. 13.** Total pressure on the structure (Top: at P1 point on the right wall; bottom: at P2 point on the bottom).

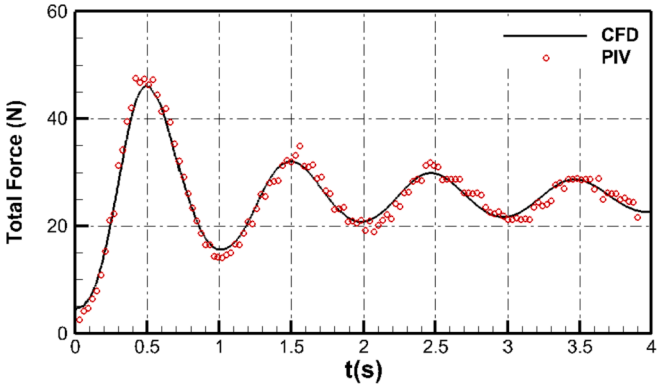



**Fig. 14.** Total force acting on the right wall of the structure.

where $K_1$ is the slope of the curve of extinction. We obtained b = 0.161 and 0.155 from the slopes ($K_1$=0.515 and 0.499) of experiment and CFD respectively. It was found that the roll damping coefficient can be estimated accurately by the numerical method having 3.7% difference with experiment in the study.

### 4.3. Estimation of velocity and vorticity fields

Especially to investigate the viscous effect on the roll motion in the roll free decay test, the velocity and vortex fields obtained from the PIV in the experiment and numerical simulations were analyzed and compared as shown in Figs. 9 ~ 12. The vorticity Ω at a point (i, j) was obtained based on the circulation calculated using its eight neighboring points, i.e. $\Omega_{i,j} = \Gamma_{i,j}/4\Delta x\Delta z$ with the circulation Γ, and Δx and Δz with the distance between adjacent velocity vectors (Jung et al., 2005).

Figs. 9 and 10 show the comparison of velocity and vorticity maps for the first negative roll angle (0.25s–0.75s), which present from the zero-down crossing to maximum negative roll angle in Fig. 7, and from the maximum negative roll angle to zero-up crossing in Fig. 10, respectively. The figures are presented when the actual roll angles are the same for accurate comparison on the velocity and vortex characteristic in vicinity of the rectangular structure.

The overall agreement for the direction, magnitude, and even extent of the vortex between experimental and CFD results was very good considering the complexity of the velocity and vorticity fields in the vicinity of the structure. When the structure was in the clockwise (CW) roll motion from t = 0.25s–0.40s as presented in Fig. 9, the strong positive vortex was seen clearly at the corner of the structure due to the CW roll motion. After the maximum negative roll angle (Fig. 10), it was observed that the positive vortex changed into the coupled vortex with a negative vortex developed at the bottom of the structure, and moved together away from the structure.

Velocity and vorticity maps from the experiment and CFD were also compared for the first positive roll angle (0.79s–1.25s) as shown in Figs. 11 and 12 including the maximum positive roll. Figs. 11 and 12 are presented at the roll angles in the CCW and CW, respectively.

When the structure was in CCW roll motion from 0.79s to 0.90s, the coupled vortex was seen clearly at the bottom of the structure, and the negative vortex detached from the main vortex as presented in Fig. 11. The detached vortex was developed into a different coupled vortex at the corner of the structure in the CW roll motion as shown in Fig. 12. The results from the CFD were also generally in good agreement with them from the PIV measurement. However, it was observed that the vortex calculated from CFD still remained strongly while the vortex obtained in experiments was getting weaker, dissipated with separation into small vortexes. The different behavior in vortex around the structure might cause the gap of the roll angle between the experimental and CFD results, especially after the two periods passed.

### 4.4. Estimation of pressure on the structure

In addition to the velocity and vorticity fields produced by the PIV technique, the pressure on the structure was obtained by computing the pressure gradients components from the flow field information. For the validation of the PIV-based estimation method, results of the unsteady total pressure at P1 on the right wall and P2 on the bottom of the structure as defined in Fig. 4 were compared with the results from the CFD simulation as shown in Fig. 13. The results from the PIV-based estimation method were in good agreement with them from the CFD simulation as expected because the static pressure was dominant comparing with the dynamic pressure in the total pressure as confirmed in Fig. 15. It was observed that the total pressure estimated at both P1 and P2 oscillated at the natural roll period as shown in Fig. 7.

Fig. 14 shows the comparison of the total force on the right wall of the structure using the predicted total pressure on the right wall. Unfortunately the total force on the bottom was not obtained because velocity fields were measured in only certain area on the bottom and changed with the roll motion of the structure in the fixed FOV as illustrated in Fig. 2. It was also observed that the total force from the PIV-based estimation method also oscillates at the natural roll period and shows good agreement with that from the CFD.

Recently there have been numerous efforts to predict the structural loads and characteristics especially for green water on offshore structures, but it is still challenging because the high-non-linear characteristics of fluid-structure interaction and violent multi-phase flow. (Ariyarathne et al., 2012; Lee et al., 2016). From the comparison of pressure between CFD and experimental results, the PIV-based pressure estimation method in the study will be alternative approach to the challenging problems with measurement using pressure gauges.

Especially the dynamic pressure obtained from the PIV-based estimation and CFD methods were analyzed and compared with static pressure to validate the methods as shown in Fig. 15. The dynamic

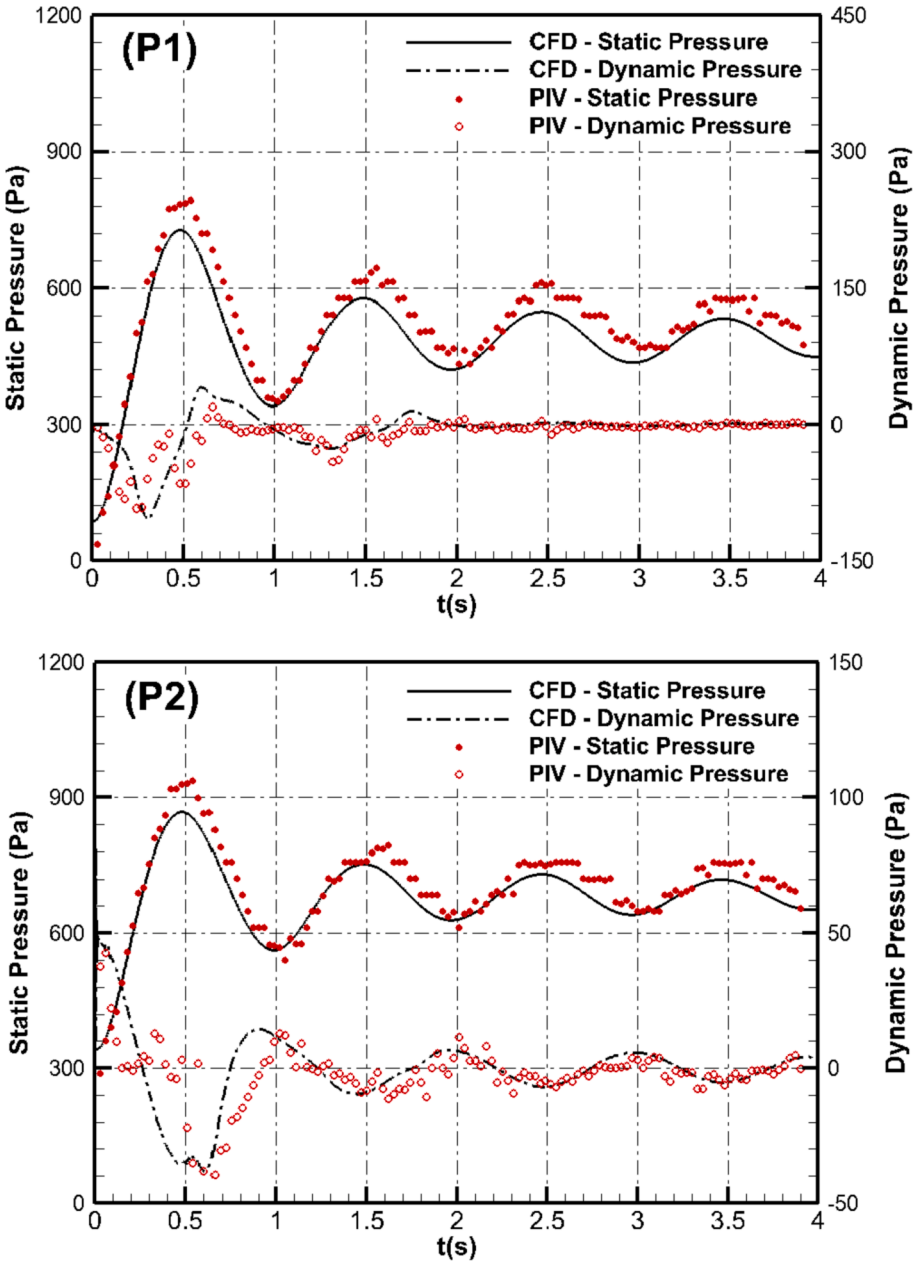



**Fig. 15.** Static and dynamic pressure on the structure (Top: at P1 point on the right wall; Bottom: at P2 point on the bottom).

pressure was obtained by subtracting the static pressure, which is determined as water depth of P1 and P2, from the total pressure. The results of the dynamic pressure from the PIV-based estimation were in general agreement with them from the CFD, even though the dynamic pressure estimated from the velocity profiles taken by PIV scattered in large roll motions. It was observed that the dynamic pressure oscillated in different way from the static pressure, and, especially at the point P2 on the bottom of the structure, in opposite way to the static pressure, even if the dynamic pressure on the bottom had the similar frequency compared to the natural roll frequency. It should be noted that the dynamic pressure on the bottom of the structure decrease the amplitude of the total pressure (e.g. static pressure is positive at all times, but dynamic pressure is negative and positive at 0.5s and 1.0s respectively). Therefore, it was found that the dynamic pressure on the bottom of the structure could play an important role in the roll damping phenomena.

It was also studied to find the vortex effect on the pressure by examining the correlation between the vortex information in the vicinity of the structure and dynamic pressure on the wall. It was deduced that the positive vortex on the bottom in Figs. 11 and 12 (i.e. 0.79s–1.25s) could have an effect on the positive dynamic pressure on the bottom of the structure as presented in Fig. 15(P2). However, it is thought that the correlation between flow information and pressure on offshore structures should be investigated further in future studies considering viscous effect influenced by hydrodynamic characteristics in a viscous shear layer region of the structures, which can be achieved by a high resolution PIV system focusing on the region.

## 5. Conclusions

Experimental and numerical studies were conducted to investigate the roll motion of 2-D rectangular structure and pressure on the structure in the free roll decay test. The experiment was performed in a 2-D wave tank with a PIV measurement to obtain velocity and vorticity maps in the vicinity of the structure. An unsteady RANS method was applied for the CFD simulations using overset grids and a forcing method to simulate the large amplitude roll motion and to minimize reflection

waves from the domain boundaries respectively. In addition to the velocity and vorticity fields obtained from the PIV technique, the PIV-based pressure estimation method was applied to predict the hydrodynamic pressure on the structure by computing the pressure gradients from the PIV velocity data based on the Euler equations. All results of the roll damping coefficient, velocity and vorticity fields, and pressure and force on the structure from the experiment and CFD were compared to validate experimental and numerical approaches applied in the studies.

The dynamic characteristics of the roll motion were investigated to predict the roll damping coefficient. The results of the roll motion from the experiment and CFD were well agreed, and the natural roll frequencies from the experiment and CFD were almost identical as 6.264 rad/s (or natural roll period $T_N = 1.003$s) and 6.274 rad/s (or $T_N = 1.001$s) respectively. The difference of the roll damping coefficients between the experiment (b = 0.161) and the CFD (b = 0.155) was 3.7%. Considering the total error of 1.466% and sampling rate of 100 Hz in the measurement of motions, the natural roll frequency and damping coefficient can be estimated accurately by the numerical method applied in the study.

The velocity and vorticity maps were analyzed to investigate the fluid fields in the roll motion in the roll free decay test. The results from the experiment and CFD were in very good agreement for the direction, magnitude and even extent of the vortex considering complexity of the fluid fields in the vicinity of the structure. The negative and positive vortex were seen clearly at the corner of the structure due to the counterclockwise and clockwise roll motions, respectively. It is found that the CFD methods applied for the numerical simulations can be employed to estimate the roll motion and viscous effect on the motion.

The pressure and loads on the structure were obtained using the PIV-based estimation method for the experiment was compared with them from the CFD to validate the numerical approach applied in the study. The total and dynamic pressure and loads from the PIV-based estimation method were in reasonable agreement with them from the numerical simulations. The total pressure estimated at the points P1 on the right wall and P2 on the bottom of the structure oscillated at the natural roll period. However, it was observed that the dynamic pressure oscillated in different way from the static pressure and even, at the P2, in opposite way to the static pressure, and the dynamic pressure on the bottom decreased the amplitude of the total pressure. It was found that the dynamic pressure could make a major contribution to the roll damping phenomena.

Overall, it is found that the CFD methods in the study can accurately estimate the roll motion with roll damping coefficient and flow fields in the vicinity of the structure in the free roll decay problem through the comparison with the experimental results. The conducted verification study of the PIV-based pressure estimation approach by comparison with the results from the CFD demonstrates the feasibility of the pressure estimation as a non-intrusive manner, and the additional capabilities of the PIV to make it an interesting alternative technique to provide similar information in a different way. It is also expected that the PIV-based estimation method would be an attractive approach to high-quality validation for CFD simulations and other scientific and engineering challenges such as green water impact on offshore structures, hydrodynamic loads for slamming on the water surface, and even problems dealing with situations when model instrumentation is problematic or impossible. Finally, the correlation between flow characteristics and pressure on offshore structures would be investigated further taking into account viscous effect on the roll motion as our future studies.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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